

Constructive Solution of the Clay Navier–Stokes Problem

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To the Board of the Clay Mathematics Institute,

1 A new form of mathematics, and a restricted version of it

On 8 September 2026 OpenAI presented a solution of alternative (C) of the Navier–Stokes Millennium problem [27]: a proof that a singular solution exists, obtained with ten thousand agents and of the order of 3×10^{11} output tokens [28]. I take it to be a serious piece of work and I do not dispute it.

What interests me first is not the result but the *kind* of mathematics that produced it. An AI is a digital optimisation process: a very large set of network parameters is fitted to given data by forward and backward propagation, the backward pass carrying the sensitivity of an output to everything that fed it. Mathematics done this way does not establish that an object exists and leave it there; it *constructs* one, by fitting. AI in this sense is a form of *automated thinking*, and the (C) solution is an instance: the singular velocity and pressure fields are built first, to have the required growth, and the force is then defined as what is left over — the residual of the construction, arranged to be smooth.

Over a long period, together with students and colleagues, I have developed a far more restricted version of the same thing: the *automated solution of differential equations*, which I will call ASDE, implemented in the FEniCS software [20, 21, 22, 23] and now widely used. ASDE fits a discrete solution to a differential equation rather than a network to data, and it does so by the same forward and backward passes. It is a small, exactly specified corner of automated thinking, and it has one advantage over the general case: what it constructs comes with a bound on the error in whatever is asked of it.

What follows is a constructive solution directed to alternative (A). I do not claim it is a solution of (A) as that alternative is formulated, and the difference should be stated at the outset rather than discovered later. (A) asks, for *all* admissible data and with $f \equiv 0$, for a solution that exists and is *smooth* for all time. What ASDE delivers, for data given in advance, is *constructive existence* and *qualitative uniqueness* of turbulent solutions. It does not deliver smoothness, because the solutions that occur do not have it — their gradients grow like $\nu^{-1/2}$ while their velocities stay bounded (Section 4).

So this is offered as an answer to what (A) is *about* — whether the equations determine a flow, and what of it they determine — and not as a discharge of (A) as written. Whether the

two coincide is for the Board to judge, and it is the one judgement I would ask for.

None of the material is new. It was published in 2007, in Chapter 14 of a book written with Johan Hoffman [35], and it has been before the Institute before. What is new is the occasion.

2 Basics of ASDE

ASDE is a self-learning computational algorithm with dual-based output error control, directed at minimal computational effort. Its three parts are the following.

A discretisation that fits the equation. Given a differential equation, a domain and data, a finite element method produces a discrete solution U whose *residual* $R(U)$ measures by how much it fails to satisfy the equation. Here the method is General Galerkin, G2: continuous piecewise linear in space and time, cG(1)cG(1), with least-squares control of the residual [15]. Nothing is supplied by the user beyond equation, domain and data — no mesh chosen by hand, no turbulence model, no closure, no constant fitted to measurement. This matters for what follows: a computation carrying a fitted turbulence model could not be used to say what the Navier–Stokes equations contain, because the turbulence would have been put in by hand.

A backward pass that converts residual into error. The error in an output $M(u) = \int_Q u \cdot \psi$ is, to leading order, the residual tested against a *dual weight function* φ ,

$$M(u) - M(U) \approx \int_0^T \int_{\Omega} R(U) \cdot \varphi \, dx \, dt, \quad |M(u) - M(U)| \leq S \|R(U)\|_{-1}, \quad (1)$$

with S the stability factor. And φ is obtained by solving the equation linearised about the computed solution *backward in time* from the output functional as data at $t = T$. This is the same object, and the same computation, as the backward pass in the training of a neural network: the adjoint of a forward evolution, propagating sensitivity from an output to everything that fed it. ASDE and AI are doing the same linear algebra in opposite directions through the same trajectory, and that is the sense in which ASDE is the restricted case.

Self-learning: adaptivity towards minimal effort. The mesh is neither fixed in advance nor refined uniformly. Refinement goes where the product of residual and dual weight is largest, and stops when the resulting bound meets the tolerance asked. Adaptivity and error control are one mechanism, not two: the computation that says how large the error is also says where to refine to reduce it, and the result is the least computational work that achieves the stated accuracy. The algorithm learns the mesh its own output requires.

What ASDE therefore produces. Not an assertion that a solution exists, but an object: the mesh with its cell count and smallest cell, the discretisation carrying the solution, the residual measured in the norm the bound uses, a computed stability factor for each output, and for each output a value with a bound on its error. That is the solution *de facto*. The record of the method and its computations is in [35], with the residual-based dissipation in [18], the parameter-free formulation in [19], the earlier statement in [26], a case at aeronautical Reynolds number — the

DLR-F11 aircraft model, time-resolved and adaptively refined — in [25], and two results the same computations gave on the way: the resolution of d’Alembert’s paradox [36] and a theory of flight [37]. Further material is collected under the label *clay problem* [40].

Where it now stands. The development since 2007 has gone to compressible flow and to aeronautical Reynolds numbers, with parallel-in-time computation of turbulent flow [24], carried substantially by Johan Jansson, who has a leading role in the AIAA CFD High Lift Prediction Workshop (HLPW-5 and HLPW-6), where computational predictions of high-lift configurations are submitted blind and compared against wind-tunnel measurement. That is the hardest ordinary test there is for a turbulence prediction: separated flow at aeronautical Reynolds number, outputs that are exactly the mean-value functionals treated below, and a comparison made before the measurement is known. ASDE is tested in the ordinary way, against experiment, by people with no stake in this letter.

3 ASDE applied to Navier–Stokes

The problem is the incompressible Navier–Stokes equations

$$\dot{u} + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad (2)$$

on a domain $\Omega \subset \mathbb{R}^3$ with initial data u_0 , forcing f and boundary conditions, at small viscosity, that is at high Reynolds number. The official statement, as set out by Charles Fefferman [31], poses them instead on $\mathbb{R}^3$ or a periodic box with the viscosity fixed, and asks whether solutions remain smooth for all time. Applied to it, ASDE delivers the following two things.

Constructive existence of turbulent solutions. A G2 computation of (2) started from smooth potential data becomes turbulent and stays bounded. The stabilisation supplies dissipation in proportion to the residual, so the dissipation is generated by the computation rather than prescribed by a model; in the vanishing-viscosity limit the same scheme is an Euler solver whose turbulent dissipation appears automatically. Existence here is not an assertion about a solution set being non-empty: a solution is exhibited, for data given in advance, with the error in each output bounded by (1). Ask a non-constructive existence theorem for the drag and there is nothing there to ask; ask a computed solution and it answers with a number and a bound.

There is more to this than error control. A computed solution is an elaborate explicit construction of a velocity–pressure field — millions of numbers on a spacetime mesh, every one available — and such an object can be taken apart: where vorticity concentrates and in what shape, how the velocity gradient decomposes at a point, where energy passes from large scales to small, where the dissipation actually sits as against where a model would put it. A solution concept whose objects can be inspected generates conjectures; one whose objects cannot be exhibited does not. The triple decomposition of the velocity gradient [16], and the energy stability analysis built on it [17], came out of looking closely at resolved velocity fields, and at measured ones [8].

Qualitative uniqueness. Classical uniqueness asks that the data determine the field pointwise. At high Reynolds number that is false, and not marginally: two computations of the same

flow on independently generated meshes give velocity fields differing by order one at any chosen point, and continuing to differ under refinement. What the data *do* determine is the qualitative content, and this is what the two computations are found to share: the drag agrees to a few per cent [38] and the lift likewise; the separation line sits in the same place; the wake is organised the same way and sheds at the same frequency; the total dissipation rate is the same.

Qualitative uniqueness is the statement that this is so, with the line between the two lists drawn *quantitatively* rather than by assertion — and that last clause is what keeps it from being a soft version of uniqueness. The dual solution attaches to each functional a computed stability factor: where it is moderate the output is determined by the data to a stated precision, and refinement where the dual weight is large will improve it; where the factor is large the output is not determined, and no refinement will make it so. The claim is falsifiable one functional at a time, and a functional can fail it. That last case is not a failure of the method but an answer: the equations do not determine that quantity, and no computation should be believed which claims otherwise.

Turbulence is a necessity, not a possibility. High Reynolds number flow starting out laminar develops shear instabilities with sharp velocity gradients, which are eventually damped by viscosity in turbulent flow. All high Reynolds number laminar flow is thus unstable and cannot continue to exist over time; it turns turbulent in order to continue to exist. What ASDE constructs is therefore not one possibility among several but what the equations must do.

4 Why smoothness is not what the solutions have

Since (A) as written asks for smoothness, it is worth being precise about what the computed solutions are instead, because they are neither smooth nor singular. Dissipation at a rate independent of the viscosity gives gradients $|\nabla u| \sim \nu^{-1/2}$, and the scales at which that gradient is realised follow from the same constant:

$$\eta \sim \left(\frac{\nu^3}{\varepsilon}\right)^{1/4} \sim \nu^{3/4}, \quad \delta u(\eta) \sim (\varepsilon \eta)^{1/3} \sim \nu^{1/4}, \quad \frac{\delta u(\eta)}{\eta} \sim \nu^{-1/2}, \quad (3)$$

the first two being the Kolmogorov–Obukhov scaling [3, 4], refined by the intermittency corrections of [5, 6] and the multifractal picture of [7], and the middle relation being Hölder continuity with exponent 1/3, the Onsager exponent. As $\nu \rightarrow 0$ the structures refine without limit while the velocity stays bounded at every scale: the gradient diverges, so the solution is not smooth, and the velocity does not, so it is not singular. A Hölder-1/3 field is exactly the object that is neither differentiable nor unbounded, which is why no amount of regularity theory reaches it.

This category is not conjectural. By the Onsager theorem [30, 34], proved after the partial results of [11, 33, 12, 13], non-smoothness is the *mechanism* by which a turbulent flow dissipates energy at a rate that does not vanish with the viscosity — the dissipation anomaly, observed since Taylor [10] and still under active examination [9] — and a local energy equation can be written in which the non-smoothness is the direct cause of the dissipation [14]. Existence of weak solutions has been available since Leray [32, 39], and for small data, the low Reynolds number case for suitably scaled problems, global existence of a smooth solution has been known for sixty years [1, 2]. The difficulty lies at the other end, and that is the end ASDE addresses.

5 The (C) solution under the viscosity test

Two observations about the announced solution, offered without disputing it.

(C) is not the negation of (A). Fefferman’s four statements do not form two pairs of negations. (A) and (B) are posed with $f \equiv 0$ and quantify over *all* admissible data; (C) and (D) ask for the *existence* of admissible data *together with an admissible smooth forcing* f for which smoothness fails. So a proof of (C) leaves (A) untouched: it does not show that a fluid tears itself into a singularity, only that something can be done to it that does. The paper states its own scope in just these terms — it establishes (C), with the periodic case giving (D) — and does not claim (A) or (B). So (A) remains open after it, which is why a solution of (A) is worth presenting.

The viscosity test. The test is general and applies to any construction exhibiting a singularity at prescribed forcing. Navier–Stokes rescales: if v solves the equations at viscosity one with forcing g , then

$$u(x, t) = a v(bx, ct), \quad a = \nu b, \quad c = ab, \quad f = a^2 b g, \quad (4)$$

solves them at viscosity ν , and the map is onto: *one* unit-viscosity blowup generates the entire family. A theorem asserting blowup “for every $\nu > 0$ ” is therefore not a family of results. It is one result, rescaled, containing exactly what its $\nu = 1$ member contains.

Everything then turns on what (4) does to the constants, and since it is the forcing class that the official statement fixes, two normalisations bear on it. Hold the solution at size one, $a = 1$: then the forcing amplitude is $a^2 b = \nu^{-1}$, so the force required *blows up* as the viscosity falls and leaves any fixed admissible class. Or hold the forcing in a fixed class, $a^2 b = 1$: then $b = \nu^{-2/3}$ and

$$\text{amplitude } a = \nu^{1/3} \rightarrow 0, \quad \text{length } b^{-1} = \nu^{2/3} \rightarrow 0, \quad \text{time } c^{-1} = \nu^{1/3} \rightarrow 0. \quad (5)$$

The singularity shrinks to nothing: at $\nu = 10^{-6}$ the velocity reaches 10^{-2} on a scale of 10^{-4} before blowing up. Neither branch survives the limit — one exits the admissible data, the other converges to the zero solution — and there is no third option. Set this beside the flow the problem is about: turbulence at small viscosity has amplitude *fixed* and structure refining, by (3). The constructed family has amplitude vanishing like $\nu^{1/3}$, which is the opposite behaviour, or else forcing diverging like ν^{-1} , which is not a fluid at all. The independent exclusion of the mechanism under analytic forcing [29] points the same way from a different direction.

6 In closing

Two constructions, then, of the same formal kind and opposite in direction. The (C) solution fixes the singularity wanted and solves backwards for a force that produces it, the force being whatever the residual requires. ASDE fixes the data a flow actually has and solves forwards, the residual becoming the bound on what may be asked of the result. Both are automated, both exhibit, and only the second exhibits a fluid.

I do not ask for the problem to be restated, and I do not claim to have discharged (A) as it is formulated. What I present is what ASDE delivers when directed at it — constructive existence

of turbulent solutions and qualitative uniqueness of their mean-value outputs, for specific data, with error bounds the computation itself produces — and I leave the Board to judge how near that comes to the question the Institute meant to ask. The material has been before the Institute over some twenty years, published from [26] onward and set out in Chapter 14 of [35]; the programme has continued since into compressible flow and aeronautical Reynolds numbers. What has changed is that a solution of (C) now exists, by methods that are themselves a new form of mathematics, and that makes it worth asking what the same kind of mathematics delivers on (A).

I would be glad to supply computational detail, or to discuss the matter with anyone the Board cares to involve.

Yours faithfully,

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